

Waffy Ahmed

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Education

Georgia Institute of Technology | Atlanta, GA
Bachelor of Science in Computer Science

December 2024
Faculty Honors, Dean's List, Zell Miller Scholar

Experience

The Home Depot

Atlanta, GA

Software Engineer II

July 2026 – Present

- Productionized a daily order-reconciliation process, replacing engineer-run scripts, laptop-initiated database tunnels, and human GCP credentials with a containerized Kubernetes workflow deployed through cdk8s, Terraform, and Spinnaker; implemented Workload Identity, least-privilege IAM, and Secret Manager/External Secrets.
- Adapted the workflow from Cloud SQL to Cloud Spanner mid-migration, engineering an in-container PGAdapter implementation, resolving cross-project IAM provisioning failures, and hardening no-data and failure paths through staged production validation before automating execution as a Kubernetes CronJob.

Software Engineer I

January 2025 – July 2026

- Increased system resilience by implementing Kubernetes Horizontal Pod Autoscaling (HPA) for a core service previously capped at 20 static pods, reducing mean latency from 121 to 72 ms and error rates by 89% while increasing throughput from 24.9M to 34.8M requests/week and enabling dynamic scaling up to 100 pods.
- Reconstructed deployment workflows for legacy Java 1.8 Tomcat services lacking operational ownership by rebuilding CI/CD pipelines and enabling zero-downtime Cassandra credential rotation, eliminating a platform outage risk capable of halting all transaction capture.
- Remediated ~330 OS-level and Python CVEs across 13 microservices by migrating Docker builds from UBI8 to a minimal internal Artifactory base image, reverse-engineering undocumented build processes and resolving CI/CD 401 authentication failures to enable secure, reproducible container builds.
- Led testing and validation to disable a legacy service component handling ~27% of transaction volume (14.5 million daily captures/lookups) using BigQuery and Postman, supporting a phased retirement effort involving 12 teams.
- Reduced mean time to recovery ~50% by implementing an automated Kubernetes rolling-restart job for 20+ microservices, replacing slow Argo Rollouts and accelerating recovery from memory-leak/state-drift issues.

Software Engineer Intern

May 2023 – July 2023, May 2024 – July 2024

- Developed an internal product information page for cashiers, streamlining self-checkout processes and reducing customer wait times for associate interventions by ~25%.
- Leveraged Java, React, and TypeScript to retrieve and surface validated product data across all self-checkout registers, mitigating customer theft by ~\$750,000 annually.
- Optimized the efficiency of configuring 40,000+ registers across 2,300 stores by 90%, in collaboration with 2 interns.
- Truncated maximum deployment time for register updates from 24+ hours to 30 minutes by using Java HttpClient and Flow APIs to streamline retrieval of polymorphic external configurations.

Projects

Fintech @ Georgia Tech | React, React Native, TypeScript, Stripe API, Chakra UI

- Reduced page load times by ~35% via transforming a multi-page credit card website into a React-based single-page application, streamlining card wallet management and adding a card removal feature for authenticated users.
- Implemented a cart page for a React Native grocery application, allowing users to add/remove items, adjust quantities, view real-time total costs, and integrated the Stripe API for streamlined in-app credit card payments.

CDC Data Reconciliation | Python, FastAPI, SQLite, React, Tailwind CSS

- Led a team of 6 to automate the reconciliation of case counts for infectious diseases between 50 state health departments and the CDC, alongside doubling as a full-stack software developer.
- Utilized Python to compare state health department and CDC data, highlighting any discrepancies between the two datasets, saving 5000+ hours of manual reconciliation annually.

Skills

Programming Languages: Java, Python, JavaScript, TypeScript, SQL

Technologies/Frameworks: Kubernetes, Google Cloud Platform (GCP), Terraform, cdk8s, Docker, Cloud Spanner, Cassandra, Elasticsearch, BigQuery, Grafana, Prometheus, OpenTelemetry, React, FastAPI, Node.js, JUnit, Git

Concepts: Object-Oriented Programming & Design, Test-Driven Development, NoSQL, Data Structures & Algorithms, Full-Stack, CI/CD, Infrastructure, Distributed Systems, Cloud Computing, Microservices Architecture